

Agentic AI for Conversational Architectural Floor Plan Generation

Tejas S. Mali¹ and Siddharth K. Gaikwad¹

¹Dept. of Computer Science and Engineering, COEP Technological University, Pune, India ,
malits24.comp@coeptech.ac.in, skg.comp@coeptech.ac.in

Abstract

Background. Architectural floor plan design is a complex and expertise-driven process that remains largely inaccessible to non-technical users during early design stages. Although recent advances in generative models and Large Language Models (LLMs) have demonstrated potential for automating layout synthesis and enabling natural language interaction, most existing systems lack autonomous decision-making and end-to-end orchestration across the design pipeline.

Methods. This study proposes an Agentic Artificial Intelligence framework for conversational architectural floor plan generation. The system integrates an LLM for requirement elicitation with a graph-based deep generative model for spatial layout synthesis. User requirements expressed in natural language are converted into structured representations, formalized as room adjacency graphs, and processed through a graph-conditioned generative model. An autonomous agent coordinates requirement extraction, graph construction, iterative refinement, and objective evaluation within a unified control loop.

Results. Experimental evaluation across multiple residential scenarios demonstrates that the proposed system consistently generates spatially coherent and topologically valid floor plans. The agent autonomously produces multiple layout variations and selects the most balanced configuration using predefined evaluation metrics, supporting rapid and accessible design exploration.

Discussion. By unifying conversational interaction, graph-based reasoning, and deep generative modeling within an agentic architecture, the proposed approach advances user-centric and accessible architectural design automation. The framework highlights the potential of Agentic AI to bridge the gap between natural language design communication and structured spatial synthesis in early-stage residential planning.

Keywords: Agentic Artificial Intelligence, Conversational Architectural Design, Floor Plan Generation, Large Language Models, Graph-Based Spatial Representation, Deep Generative Models

1 Introduction

Architectural floor plan design forms the foundation of residential building development, directly influencing spatial efficiency, circulation flow, usability, environmental comfort, and long-term adaptability. In early-stage design, architects translate abstract client requirements into structured spatial configurations through iterative sketching, constraint negotiation, and refinement. This process requires substantial domain expertise, spatial reasoning skills, and repeated client–architect interaction. Consequently, early conceptual design remains largely inaccessible to non-technical stakeholders, who may find it difficult to communicate spatial expectations in formal architectural terms. These challenges motivate the development of intelligent computational systems capable of assisting or automating layout generation while preserving design coherence and functional validity.

Over the past decade, advances in data-driven design and deep generative modeling have significantly transformed architectural layout synthesis. Graph-based representations have been widely adopted to encode room relationships and adjacency constraints, enabling structured spatial reasoning (Hu et al., 2020). Data-centric frameworks such as RPLAN have demonstrated how large annotated datasets can support learning-based interior plan generation (Wu et al., 2019). Generative Adversarial Networks (GANs), particularly iterative refinement architectures such as House-GAN++, further improved spatial realism by enforcing relational consistency among rooms (Nauata et al., 2021). Diffusion-based approaches, including vectorized denoising models, have enhanced diversity and geometric continuity in floor plan synthesis (Shabani et al., 2022). While these methods achieve strong generative performance, they typically assume well-defined graph inputs or manually structured constraints, thereby limiting direct usability for non-expert users.

In parallel, the emergence of Large Language Models (LLMs) has enabled natural language-driven interaction across diverse design domains. Conversational systems have demonstrated the ability to interpret user intent, translate textual descriptions into structured representations, and facilitate intuitive human–AI collaboration (Li et al., 2024), (Lin et al., 2023). In architectural contexts, such models reduce the technical barrier for communicating design requirements and support client-centric engagement. However, existing language-guided layout systems often treat the LLM as an isolated translation module rather than as an autonomous reasoning component integrated throughout the pipeline. As a result, downstream generation, evaluation, and refinement stages frequently remain detached from conversational intelligence.

Beyond isolated generation techniques, the broader concept of Agentic Artificial Intelligence introduces a paradigm in which systems autonomously plan, reason, evaluate, and adapt toward achieving a defined objective (Park et al., 2023). Agentic systems maintain internal state representations, coordinate multi-stage workflows, and iteratively refine outputs without continuous human intervention. In architectural computation, reinforcement learning and agent-based frameworks have been explored for spatial optimization and generative concept discovery (Kakooee and Dillenburger, 2025), (Zhang et al., 2024). Nevertheless, a unified framework that combines conversational requirement elicitation, graph-based spatial reasoning, deep generative synthesis, and autonomous evaluation within a single agentic loop remains largely unexplored.

The absence of such integration creates a critical research gap. Existing systems either excel at geometric generation but lack accessibility, or enable natural language interaction without robust autonomous reasoning. Furthermore, many approaches require manual model selection, parameter tuning, or post-generation comparison, preventing true end-to-end automation. For early-stage residential design to become widely accessible, computational systems must not only generate layouts but also understand user intent, translate requirements into structured constraints, evaluate multiple alternatives, and autonomously select optimal configurations.

To address these limitations, this paper proposes an Agentic AI framework for conversational architectural floor plan generation. The system transforms natural language input into structured design specifications through iterative conversational reasoning. Extracted requirements are formalized as a relational room graph, which serves as input to a graph-conditioned deep generative model for spatial synthesis. Multiple layout variations are produced and evaluated using objective criteria reflecting spatial balance and topological consistency. A centralized design agent orchestrates these stages through an autonomous control loop, maintaining system state and performing self-evaluation to select the most

coherent layout.

The primary contribution of this work lies in unifying conversational interaction, symbolic graph reasoning, and neural generative modeling within a single agentic architecture. Unlike prior systems that treat language understanding and layout generation as separate modules, the proposed framework integrates them into a cohesive multi-stage reasoning pipeline. This integration enables fully automated early-stage residential layout generation while preserving interpretability, structural validity, and user accessibility.

By advancing an agentic approach to architectural floor plan synthesis, this research aims to bridge the gap between human-centered design communication and data-driven generative modeling. The proposed framework supports rapid design exploration, reduces iteration time, and enhances accessibility for non-technical users, contributing toward more inclusive and intelligent computer-aided architectural design systems.

2 Related Work

Research on automated architectural layout generation has evolved across multiple computational paradigms, including graph-based modeling, deep generative learning, reinforcement learning, and natural language-driven interaction. This section reviews prior work most relevant to floor plan synthesis and situates the proposed approach within existing literature.

2.1 Graph-Based and Data-Driven Floor Plan Generation

Early data-driven approaches to residential layout generation represent floor plans as structured graphs encoding room types and adjacency relationships. Graph2Plan learns a mapping from layout graphs to geometric floor plans, demonstrating that topological constraints can guide spatial synthesis (Hu et al., 2020). Similarly, RPLAN introduced a large-scale dataset of annotated residential layouts and highlighted the importance of graph-structured representations for realistic plan generation (Wu et al., 2019). Multimodal approaches further integrate topology and spatial semantics to improve layout expressiveness (El Mesawy et al., 2024). Large-scale annotated datasets such as CubiCasa5k have also supported advances in floor plan understanding and generation tasks (Kalervo et al., 2019). Subsequent work extended graph-constrained formulations to improve spatial feasibility and room connectivity (Aalaei et al., 2023). While these methods provide structured control, they generally assume graph inputs provided by experts, limiting usability for non-technical users.

2.2 Deep Generative Models for Layout Synthesis

GANs have been widely adopted for floor plan generation due to their ability to model complex spatial distributions. House-GAN++ introduced an iterative layout refinement framework using convolutional message passing to enforce adjacency constraints during generation (Nauata et al., 2021). This approach significantly improved layout coherence compared to earlier one-shot GAN models. More recent diffusion-based methods, such as HouseDiffusion, model layout generation as a denoising process over vector representations, yielding smoother room boundaries and improved diversity (Shabani et al., 2022). Hybrid approaches combining GANs with evolutionary optimization have also been explored to balance design quality and constraint satisfaction (Pham and Nguyen, 2025). Optimization-driven studies further demonstrate how architectural layout and environmental constraints can be jointly considered in spatial planning (Chen et al., 2025). Despite their effectiveness, these models typically operate as standalone generators and lack integrated mechanisms for requirement interpretation and autonomous evaluation.

2.3 Conversational and Language-Guided Design Systems

The emergence of LLMs has enabled natural language interaction for early-stage design tasks. Chat-Design demonstrated that language models can bootstrap floor plan generation by translating textual descriptions into structured layout representations (Li et al., 2024). Prompt-guided layout generation frameworks further explore how textual cues can influence spatial arrangements (Qin et al., 2024), (Lin et al., 2023). Beyond layout synthesis, conversational interfaces have also been explored to enhance architect-client communication and early design collaboration in BIM-enabled workflows (Bagasi and Nawari, 2025). These systems reduce the technical barrier for users but often rely on predefined pipelines where language models serve primarily as input translators rather than autonomous decision makers.

2.4 Agentic and Autonomous Design Frameworks

Agentic AI emphasizes goal-driven behavior, multi-step reasoning, and autonomous decision-making (Park et al., 2023). In architectural contexts, agent-based and reinforcement learning methods have been applied to spatial optimization, generative concept discovery, and robotic construction planning (Kakooee and Dillenburger, 2025), (Zhang et al., 2024), (Tsumoto et al., 2025), (Leder and Menges, 2024). However, such systems typically focus on optimization within constrained environments and do not integrate conversational interaction or end-to-end generative pipelines. Surveys of AI techniques in early architectural design further highlight the lack of unified agentic frameworks that combine interaction, generation, and evaluation (Vissers-Similon et al., 2025), (Yan et al., 2025).

2.5 Positioning of the Proposed Work

In contrast to prior approaches, the proposed system integrates conversational language understanding, graph-based generative modeling, and autonomous evaluation within a single agentic control loop. Rather than treating language models, layout generators, and evaluation mechanisms as isolated components, the system orchestrates them through multi-stage reasoning to achieve end-to-end automation of residential floor plan design. This integration addresses limitations of existing methods related to accessibility, autonomy, and iterative decision-making.

3 Proposed Methodology

This section describes the proposed Agentic AI framework for conversational architectural floor plan generation. The methodology is designed as an end-to-end autonomous pipeline that transforms natural language user input into a complete residential layout through multi-stage reasoning and decision-making. The system integrates conversational language understanding, graph-based representation, deep generative modeling, and automatic evaluation under a unified agentic control loop.

3.1 Overall System Architecture

The proposed framework follows a modular architecture coordinated by a central design agent. The agent maintains the system state, orchestrates data flow between components, and autonomously decides execution strategies at each stage of the pipeline. An overview of the system architecture is illustrated in Fig. 1.

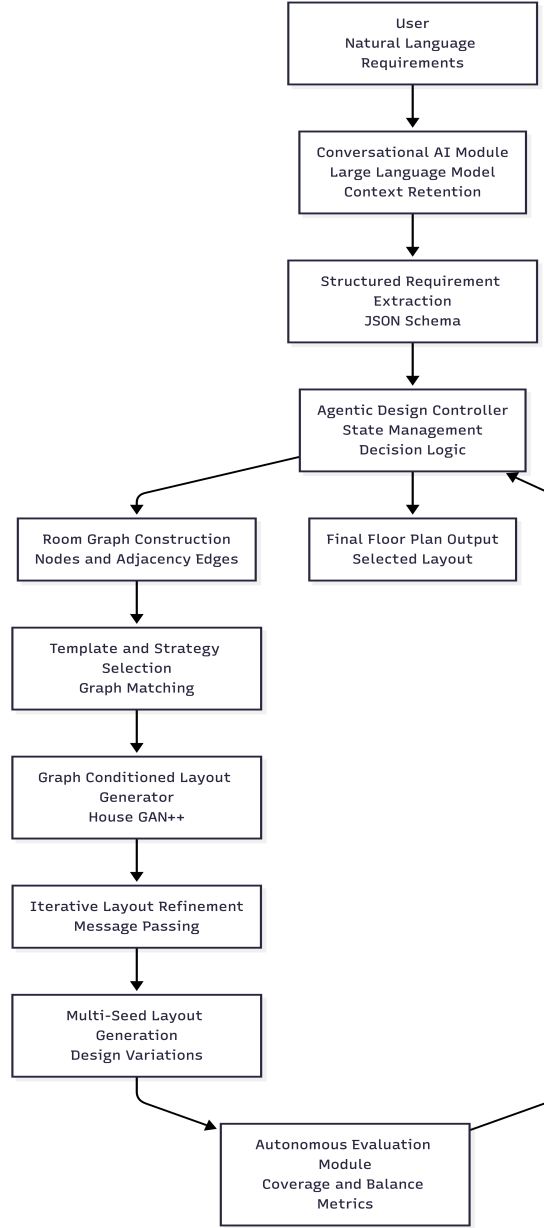


Figure 1: Overall architecture of the proposed agentic floor plan generation framework.

The pipeline consists of five primary stages: conversational requirement elicitation, structured requirement extraction, room graph construction, layout generation using a deep generative model, and autonomous evaluation with layout selection.

3.2 Conversational Requirement Elicitation

User interaction with the system is initiated through natural language dialogue. An LLM is employed to interpret user intent, ask clarifying questions when necessary, and accumulate contextual information across multiple conversation turns. This conversational interaction paradigm is consistent with prior

work on natural language-driven human-system interaction in construction and design contexts (Li et al., 2024), (Park et al., 2024), (Lin et al., 2023). Unlike prompt-only systems, the conversational module supports incremental clarification and context retention, enabling more accurate capture of user requirements.

3.3 Structured Requirement Extraction

Once sufficient conversational context is obtained, the agent extracts structured design requirements in a predefined schema. These requirements include room counts, optional spaces, and high-level layout preferences. Structured extraction enables downstream processing using graph-based representations, which are commonly adopted in architectural layout generation research (Wu et al., 2019), (Hu et al., 2020). This step allows the system to bridge unstructured language input and formal spatial representations without manual intervention.

3.4 Room Graph Construction

Based on the extracted requirements, the agent constructs a relational graph $G = (V, E)$, where V represents the set of room nodes and E represents the set of adjacency edges. Each node $v_i \in V$ is assigned a semantic label $l_i \in L$ (e.g., Bedroom, Kitchen) from a predefined set of 19 room types. The adjacency matrix A is constructed such that $A_{ij} = 1$ if room i and room j share a boundary, and 0 otherwise. This graph-based representation captures topological constraints essential for residential layouts (Aalaei et al., 2023), (Hu et al., 2020). The construction process incorporates domain-informed adjacency rules to ensure spatial feasibility.

3.5 Deep Generative Layout Synthesis

The generated room graph is provided as input to a graph-conditioned deep generative model for layout synthesis. In this work, a Generative Adversarial framework with iterative refinement is employed to produce spatially coherent room masks from the input graph (Nauata et al., 2021). Iterative refinement allows the model to progressively adjust room boundaries while respecting adjacency constraints, leading to improved layout consistency compared to single-pass generation methods.

3.6 Autonomous Evaluation and Selection

To support design exploration, the agent generates multiple layout variations for a single set of user requirements. To select the optimal result without human intervention, the system employs an objective ranking mechanism. Each generated variation is evaluated based on two primary factors: *Spatial Balance*, which favors proportional room area distributions, and *Connectivity Coverage*, which ensures that generated room masks respect the topological edges defined in the input graph. The layout demonstrating the highest adherence to these spatial constraints is autonomously selected by the agent and presented to the user.

3.7 Agentic Control Loop

The defining characteristic of the proposed methodology is the agentic control loop that coordinates all stages of the pipeline. Inspired by agent-based AI principles (Park et al., 2023), the agent maintains internal state, evaluates intermediate results, and adapts execution flow without external supervision. This design enables true end-to-end automation of early-stage floor plan generation rather than isolated model inference.

4 Experiments and Results

This section evaluates the proposed agentic framework through a set of controlled experiments designed to assess layout quality, spatial coherence, and autonomous decision-making. The experiments focus on residential floor plan generation scenarios commonly studied in prior work, enabling qualitative and quantitative comparison with existing approaches.

4.1 Experimental Setup

The system is evaluated on multiple residential design scenarios, including one-bedroom, two-bedroom, and three-bedroom configurations with varying optional spaces such as balconies or study rooms. These configurations align with layout distributions commonly observed in publicly available residential datasets and prior floor plan generation studies (Wu et al., 2019), (Nauata et al., 2021).

The deep generative component is based on a graph-conditioned generative adversarial model trained on residential layout data. For each user requirement set, the agent generates multiple layout variations by sampling different random seeds. This strategy follows established practices in generative layout modeling to explore spatial diversity (Shabani et al., 2022). All experiments are conducted using the same model configuration to ensure consistency across scenarios.

4.2 Evaluation Metrics

Evaluating architectural layouts remains a challenging task due to the absence of universally accepted quantitative metrics. In line with prior research, this work adopts objective spatial metrics that approximate layout quality without relying on subjective human judgment (Nauata et al., 2021), (Hu et al., 2020). The evaluation focuses on:

- **Room Coverage:** Verifying that generated layouts avoid fragmented or missing spaces.
- **Area Balance:** Assessing the distribution of room areas to prevent extreme size imbalance (e.g., excessively large corridors).
- **Constraint Satisfaction:** Verifying adherence to basic adjacency constraints encoded in the room graph.

4.3 Qualitative Results

Fig. 2 illustrates representative floor plans generated by the proposed system for different residential scenarios. The generated layouts demonstrate coherent spatial organization, clear room separation, and adherence to expected residential adjacency patterns, such as connectivity between living spaces and entrances or proximity between bedrooms and bathrooms. These qualitative observations are consistent with findings reported in graph-based and GAN-based layout generation literature (Nauata et al., 2021), (Hu et al., 2020).

4.4 Performance Evaluation

Table 1 summarizes the system’s performance across different complexity levels. The results indicate that the system can generate valid, multi-room layouts in under one minute on standard hardware. The "Graph Complexity" metric reflects the number of nodes (rooms) and edges (connections) managed by the agent, demonstrating the system’s ability to handle increasingly complex spatial requirements while maintaining reasonable inference times.

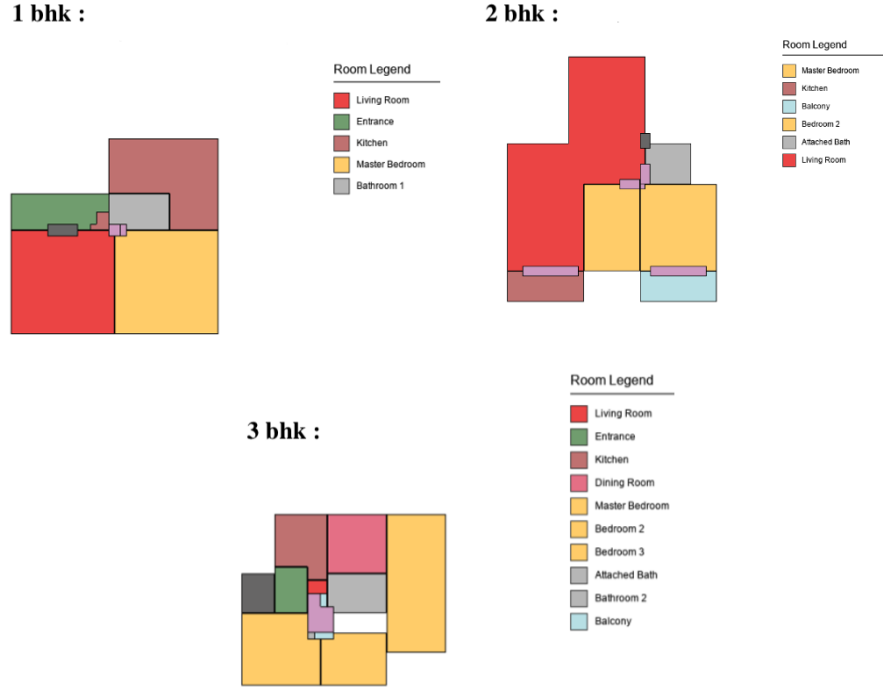


Figure 2: Example floor plans generated by the proposed system for different residential configurations.

Table 1: System Performance Across Residential Scenarios

Scenario	Graph Complexity	Variations Gen.	Avg. Gen. Time
1 BHK	4-5 Nodes	6	≈ 45 sec
2 BHK	6-7 Nodes	6	≈ 55 sec
3 BHK	8-10 Nodes	6	≈ 65 sec

4.5 Discussion

The experimental results indicate that integrating conversational input, graph-based representations, and deep generative modeling within an agentic framework enables consistent and coherent floor plan generation. Unlike prior systems that require manual intervention at intermediate stages, the proposed approach demonstrates fully autonomous end-to-end operation. While quantitative evaluation remains limited by the lack of standardized architectural metrics, the observed results align with established benchmarks reported in related generative layout studies (Yan et al., 2025), (Shabani et al., 2022).

5 Limitations and Ethical Considerations

While the proposed agentic framework demonstrates promising results in conversational architectural floor plan generation, several limitations remain. First, the system operates primarily at a topological and relational level, focusing on room adjacency and proportional balance rather than precise geometric dimensions or regulatory compliance. As a result, the generated layouts should be interpreted as conceptual design suggestions rather than construction-ready blueprints. Real-world implementation would

require integration with building codes, structural constraints, ventilation requirements, and local zoning regulations.

Second, the generative model is trained on existing residential layout datasets, which may reflect implicit design biases present in the source data. These biases could influence room arrangements, spatial hierarchies, or cultural design patterns embedded within the generated outputs. Although the agent autonomously evaluates spatial balance and connectivity, it does not explicitly reason about socio-cultural diversity, accessibility standards, or sustainability criteria.

From an ethical perspective, the deployment of automated design systems raises important considerations regarding professional responsibility and authorship. The framework is intended to assist early-stage conceptual exploration rather than replace professional architects. Human expertise remains essential for validating safety, functionality, and contextual appropriateness. Additionally, transparency in how design decisions are derived—particularly in graph construction and layout selection—should be maintained to ensure interpretability and user trust.

Future research should address these limitations by incorporating regulatory knowledge bases, fairness-aware dataset curation, and explainable decision mechanisms within the agentic control loop. Such enhancements would further strengthen the responsible application of Agentic AI in architectural design workflows.

6 Conclusion and Future Work

This paper presented an Agentic AI framework for conversational architectural floor plan generation that enables end-to-end automation of early-stage residential layout design. By integrating conversational language understanding, graph-based spatial representation, and deep generative modeling within an autonomous control loop, the proposed system transforms natural language user requirements into coherent floor plans without requiring expert intervention. The experimental evaluation demonstrates that the framework consistently produces spatially structured layouts across multiple residential scenarios while supporting autonomous variation generation and selection.

Unlike prior approaches that rely on predefined graphs, manual parameter tuning, or isolated model inference, the proposed method emphasizes agentic behavior through multi-stage reasoning, self-evaluation, and decision-making. This design aligns with emerging perspectives on agent-based AI systems that operate toward a goal through autonomous planning and execution ([Park et al., 2023](#)). The results indicate that combining graph-conditioned generative models with conversational interfaces can significantly improve accessibility and usability in architectural design workflows, complementing existing data-driven layout generation methods ([Nauata et al., 2021](#)), ([Hu et al., 2020](#)).

Despite these strengths, the current system has several limitations that motivate future research. First, the generated layouts do not explicitly incorporate real-world dimensional constraints or regulatory requirements, which are essential for construction-ready designs. Second, the framework currently focuses on single-floor residential layouts and does not address multi-story structures or vertical circulation. Third, quantitative evaluation remains constrained by the lack of standardized metrics for architectural layout quality, a limitation widely acknowledged in prior studies ([Yan et al., 2025](#)).

Future work will explore the integration of explicit geometric constraints, multi-floor layout generation, and interactive refinement based on user feedback. Additionally, extending the agentic framework to support style preferences, regulatory compliance, and interoperability with Building Information Modeling

(BIM) tools represents a promising direction. These extensions aim to further bridge the gap between early-stage conceptual design and practical architectural workflows, advancing the role of Agentic AI in computer-aided architectural design.

References

- M. R. Aalaei, M. Saadi, M. Rahbar, and A. Ekhlassi. Architectural layout generation using a graph-constrained conditional generative adversarial network (gan). *Automation in Construction*, 155: 105053, 2023.
- O. Bagasi and N. O. Nawari. Bim and ai in early design stage: Advancing architect-client communication. *Buildings*, 15(12):1977, 2025.
- X. Chen, H. Kang, J. Zhao, and Q. Liu. Optimization design research of architectural layout and morphology in multi-story dormitory areas based on wind environment analysis. *Buildings*, 15(10): 1747, 2025.
- M. El Mesawy, N. A. Zaher, and A. El Antably. From topology to spatial information: A computational approach for generating residential floorplans. In *Proceedings of the 29th International Conference on Computer-Aided Architectural Design Research in Asia (CAADRIA)*, pages 129–138, 2024.
- R. Hu, Z. Huang, Y. Tang, O. van Kaick, H. Zhang, and H. Huang. Graph2plan: Learning floorplan generation from layout graphs. *ACM Transactions on Graphics (TOG)*, 39(4):118, 2020.
- R. Kakooee and B. Dillenburger. Enhancing architectural space layout design by pretraining deep reinforcement learning agents. *Journal of Computational Design and Engineering*, 12(1):149–166, 2025.
- A. Kalervo, J. Ylioinas, M. Häikiö, A. Karhu, and J. Kannala. Cubicasa5k: A dataset and an improved multi-task model for floorplan image analysis. *arXiv preprint arXiv:1904.01920*, 2019.
- S. Leder and A. Menges. Merging architectural design and robotic planning using interactive agent-based modelling for collective robotic construction. *Journal of Computational Design and Engineering*, 11(2):253–268, 2024.
- J. Li, Y. Luo, S. Lu, J. Zhang, J. Wang, R. Guo, and S. Wang. Chatdesign: Bootstrapping generative floor plan design with pre-trained large language models. In *Proceedings of the 29th International Conference on Computer-Aided Architectural Design Research in Asia (CAADRIA)*, pages 99–108, 2024.
- J. Lin, J. Guo, S. Sun, Z. Yang, J.-G. Lou, and D. Zhang. Layoutprompter: Awaken the design ability of large language models. In *Proceedings of the Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 43852–43879, 2023.
- N. Nauata, S. Hosseini, K.-H. Chang, H. Chu, C.-Y. Cheng, and Y. Furukawa. House-gan++: Generative adversarial layout refinement network towards intelligent computational agent for professional architects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16230–16240, 2021.

- S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 1–22, 2023.
- S. Park, X. Wang, C. Menassa, V. Kamat, and J. Chai. Natural language instructions for intuitive human interaction with robotic assistants in field construction work. *Automation in Construction*, 161:105345, 2024.
- V. H. S. Pham and D. H. Nguyen. A hybrid approach to apartment layout planning using generative adversarial networks and genetic algorithms. *Engineering, Construction and Architectural Management*, 32(9), 2025.
- S. Qin et al. Chathousediffusion: Prompt-guided generation and editing of floor plans. *arXiv preprint arXiv:2410.11908*, 2024.
- M. A. Shabani, S. Hosseini, and Y. Furukawa. Housediffusion: Vector floorplan generation via a diffusion model with discrete and continuous denoising. *arXiv preprint arXiv:2211.13287*, 2022.
- R. Tsumoto, K. Yaji, Y. Nomaguchi, and K. Fujita. Deep concept identification for generative design. *Advanced Engineering Informatics*, 65:103354, 2025.
- E. Vissers-Similon, T. Dounas, and J. De Walsche. Classification of artificial intelligence techniques for early architectural design stages. *International Journal of Architectural Computing*, 23(2):387–404, 2025.
- W. Wu, X.-M. Fu, R. Tang, Y. Wang, Y.-H. Qi, and L. Liu. Data-driven interior plan generation for residential buildings. *ACM Transactions on Graphics (TOG)*, 38(6):234, 2019.
- S. Yan, C. Wu, and Y. Zhang. Generative design for architectural spatial layouts: A review of technical approaches. *Journal of Asian Architecture and Building Engineering*, 2025. Published online Jun. 2025.
- Z. Zhang, Z. Guo, H. Zheng, and Z. Li. Automated architectural spatial composition via multi-agent deep reinforcement learning for building renovation. *Automation in Construction*, 167:105702, 2024.